

CHESTER COUNTY AP, PA, USA

WMO: 720408

Lat: 39.983N

Lon: 75.867W

Elev: 201

StdP: 98.93

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 00136

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.2	-9.9	-20.0	0.6	-10.4	-17.5	0.8	-8.0	12.3	-3.1	11.4	-3.0	4.3	300	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	32.2	24.5	30.8	23.8	28.9	22.8	25.9	29.6	24.9	28.6	24.0	27.3	2.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.5	28.0	23.8	19.1	26.6	22.8	18.0	25.4	81.8	29.2	76.4	29.3	73.2	26.9	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.5	8.8	7.5	-14.8	34.1	2.9	2.2	-16.8	35.7	-18.5	36.9	-20.2	38.2	-22.3	39.8		
			-15.8	27.7	2.6	2.1	-17.6	29.2	-19.2	30.4	-20.6	31.6	-22.5	33.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	-0.8	0.9	4.9	11.5	16.9	21.3	23.9	22.5	19.4	13.2	6.7	2.6
	DBStd	9.58	5.32	5.47	5.13	4.94	4.39	3.17	2.75	2.54	3.68	4.57	4.57	4.81
	HDD10.0	1181	336	258	173	38	2	0	0	0	0	19	118	236
	HDD18.3	2841	593	487	417	213	81	11	0	1	32	169	349	488
	CDD10.0	1904	1	4	16	83	218	338	432	387	281	119	19	6
	CDD18.3	520	0	0	0	8	38	98	174	129	63	11	0	0
	CDH23.3	3777	0	0	3	71	280	719	1432	839	386	47	0	0
	CDH26.7	1061	0	0	0	13	68	184	483	202	103	6	0	0
Wind	WSAvg	2.8	3.4	3.5	3.6	3.3	2.6	2.2	1.8	1.8	2.0	2.6	3.0	3.2
Precipitation	PrecAvg	1191	94	65	113	97	88	114	104	102	123	102	86	100
	PrecMax	1634	155	147	194	187	179	287	199	196	326	264	127	205
	PrecMin	881	41	12	12	42	36	48	29	41	10	18	26	24
	PrecStd	209	34	28	43	39	37	66	46	39	81	64	31	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.1	19.8	22.7	28.7	31.1	32.6	34.1	32.2	32.1	27.3	20.9	17.8
		MCWB	13.6	13.8	15.4	17.4	23.1	24.5	25.4	24.5	24.0	20.7	15.6	16.6
	2%	DB	12.4	15.0	18.7	25.9	28.7	30.9	32.4	30.1	29.8	24.1	17.9	15.0
		MCWB	10.6	11.3	13.8	17.5	22.3	23.3	24.9	23.8	22.5	19.8	14.3	14.2
	5%	DB	9.0	12.1	16.0	22.5	27.0	28.7	31.1	28.8	27.5	22.3	16.1	12.6
		MCWB	7.3	8.5	11.3	16.4	20.6	22.6	24.3	23.1	21.5	19.0	12.9	11.5
	10%	DB	7.0	8.0	12.8	20.0	24.1	27.4	29.0	27.6	25.3	20.2	13.8	9.8
		MCWB	5.1	5.3	9.3	14.6	19.0	21.3	23.3	22.1	20.1	17.5	10.8	8.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.1	15.1	16.9	22.2	25.8	27.1	28.1	26.7	25.2	22.9	17.9	17.3
		MCDB	14.2	18.0	20.4	24.0	28.7	28.8	29.8	29.4	29.6	24.2	18.6	17.5
	2%	WB	11.1	11.6	14.5	19.0	22.9	24.6	26.3	25.3	23.9	21.3	15.6	14.2
		MCDB	12.0	13.7	17.4	22.8	26.0	28.7	30.3	28.9	27.5	23.4	17.2	15.0
	5%	WB	7.7	7.9	12.4	17.0	21.8	23.5	25.1	24.2	22.9	19.8	13.7	12.1
		MCDB	8.5	10.0	15.0	21.4	25.5	27.1	29.1	26.9	25.5	21.7	15.4	12.7
	10%	WB	5.1	5.8	9.9	15.3	20.5	22.5	24.2	23.3	22.1	17.8	12.0	8.4
		MCDB	6.7	8.0	12.9	19.2	23.4	25.8	27.8	25.8	24.5	19.2	13.6	9.1
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.6	8.7	10.2	9.6	9.4	9.2	8.8	9.1	8.7	8.2	6.6
		MCDBR	10.7	11.6	12.8	12.9	10.9	10.8	10.3	9.6	10.4	10.0	10.5	9.2
		MCWBR	9.6	9.1	8.8	7.2	6.1	5.2	4.9	4.4	4.9	5.8	7.4	8.8
	5% WB	MCDBR	9.9	10.5	12.1	11.9	10.2	9.6	9.2	8.3	8.9	8.3	9.0	9.1
		MCWBR	9.6	9.1	9.7	7.9	6.2	5.5	5.0	4.6	4.9	5.9	8.0	9.3
Clear-Sky Solar Irradiance	taub	0.305	0.316	0.351	0.389	0.430	0.493	0.496	0.495	0.409	0.355	0.319	0.302	
	taud	2.455	2.404	2.379	2.293	2.212	2.074	2.081	2.069	2.281	2.446	2.510	2.504	
	Ebn at Noon	863	902	902	885	852	796	789	777	829	846	840	839	
	Edh at Noon	79	96	109	128	142	163	160	158	120	91	75	70	
All-Sky Solar Radiation	RadAvg	1.90	2.79	3.69	4.79	5.36	5.88	5.96	5.21	4.29	3.07	2.18	1.64	
	RadStd	0.16	0.33	0.28	0.42	0.65	0.47	0.29	0.32	0.52	0.33	0.19	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (79 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+126	+73

Nomenclature: See separate page